

General Features

The i305 is a CAN, RS485/422 and RS232 interface card for the ionia™ system. The module includes two galvanically isolated interfaces which can be individually selected as CAN, RS485/422 or RS232.

With the i305 a gateway to the consist network (being either on the central unit or on any other module) can easily be realised. Moreover, any other devices can be connected to the system and thus can be controlled out of one single application.

Both CAN 2.0 controllers (fully compliant with ISO11898), and the UART controllers are implemented in a FPGA with proven duagon IP. The CANopen master and slave functions as well as the user-defined RS485 bus administrators are implemented in the FPGA, by the use of a 32bit soft-core CPU.

The configuration registers and all communication data are accessible by the central unit through a fast and secure point-to-point connection (duagon serial star slave).

A built-in supervisory mechanism called «Condition Monitoring» can detect, record and report any off-limit usage, such as overvoltage, overcurrent, supply interruption, overtemperature, etc. as well as selected user-defined values.

Configuring the i305 is done easily and time saving by using one of the five methods defined by IEC 61131.

The i305 is designed for harsh rolling stock environments and is fully compliant to EN 50155 and IEC 61375.

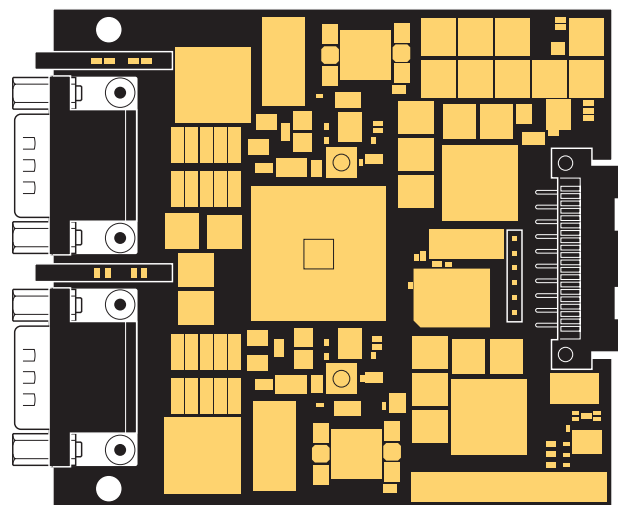
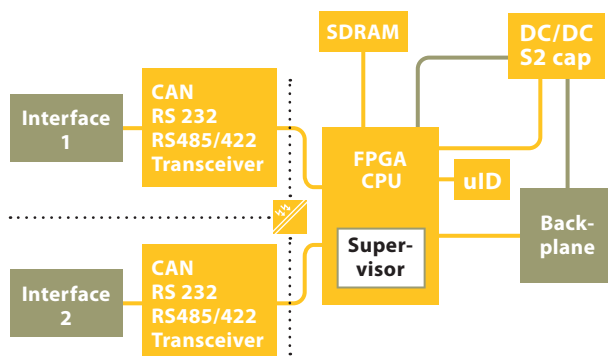
Key Benefits

- CAN 2.0 controllers fully compliant to ISO 11898 standard
- CANopen Master and Slave communication stacks running on the module, no additional CPU power needed.
- Easy access to the CANopen communication by a graphical editor
- Configurable UART controller manages the connection to any other devices.
- Fully integrable into the central units application
- No obsolescence risk due to the use of duagon® own soft-IP in an FPGA

Application Examples

- CAN and/or RSxxx Gateway to main consist network (in combination with central unit)
- Gateway to any other field bus (in combination with other communication modules)
- Gateways between CAN and RSxxx without other modules
- Connection to single devices with other fieldbuses
- CAN to CANopen gateway / protocol translation

i305 Hardware Architecture



Life Cycle Cost

The use of an FPGA with included duagon own soft-IP minimizes the risk and costs in case of component obsolescence. The i305 has been strictly designed without electrolytic capacitors to avoid service costs for changing capacitors.

Any special application (e.g. a custom specific bus master for RS485) can be programmed by the customer itself or by duagon®.

Technical Data

CAN	- Two galvanically isolated CAN channels with standard CiA DS-102 pin assignment - CAN 2.0a/b compliant to ISO 11898 (tested according to ISO 16845) - CAN pie 2.0 - CANopen master or slave (NMT) compliant with IEC 61375 - J1939 as option - Configurable baud rate: up to 1 MBaud 120 Ω or 150 Ω termination selectable	Backplane Communication	100 Mbit duagon serial star slave - Secure communication protocol
RS485/RS422/RS232	- Two galvanically isolated RS485/RS422/RS232 channels - Configurable baud rate: 9600 up to 115 kBaud - Configurable data format (parity, stopbit, data length) 0.75 V bias voltage selectable 120 Ω or 150 Ω termination selectable	Diagnostic/Service	- LEDs for visual status information (power on, running, communicating, failure, I/O, ...) - Diagnostic information readable through the central unit's service interfaces (see i101/i102/i103 for further information about these possibilities)
Programming	- Programmable by the central unit using IEC 61131 LD, FBD, ST, IL and SFC - Firmware update handled by the central unit	Power Consumption	$P_{\max} < 5 \text{ W}$; $P_{\text{save}} < 1 \text{ W}$ - Supports sleep mode - Power supply via backplane - Interruption class S2 with ceramic capacitors
Security / Watchdog	- External supervisor for voltage, current and temperature monitoring → «Condition Monitoring» - Communication Link with watchdog supervision	Operating Conditions	- Ambient temperature: -40 to +85 °C (EN 50155, Class TX) - Relative humidity: 75%, max 95% for 30 days per year (conformal coating) EN 60068 - Shock and vibration: - According to IEC 61373 category 1, class B - EMI: - According to EN 50121 and EN 50155
		Dimensions/Weight	4TE × 92 × 86 mm, fitting the ionia™ housing 110 g
		Environment	Fully compliant to RoHS and REACH

d-006368-013035

Order Code and HW Options

Product Type: i305 - - - - - - - -													
Interface 2: optional													
EXAMPLES: i305-i1R4-UD-12B-U i305-i1C-OM-12-i2R2-U													
Interface 1		Protocol		Termination & Bias		Interface 2		Protocol		Termination & Bias		DSUB Bolt	
CAN	i1C	CAN	L	120 Ω	12	CAN	i2C	CAN	L	120 Ω	12	UNC	U
		CANopen Master	OM	150 Ω	15			CANopen Master	OM	150 Ω	15		
		CANopen Slave	OS					CANopen Slave	OS				
		J1939	J					J1939	J				
RS485/ RS422	i1R4	Uart p to p	UA	120 Ω Term	12	RS485/ RS422	i2R4	Uart p to p	UA	120 Ω Term	12		
		User-defined*	UD	150 Ω Term	15			User-defined*	UD	150 Ω Term	15		
				0.75 V Bias	B					0.75 V Bias	B		
RS232	i1R2					RS232	i2R2						

Contact duagon for further information about this possibilities

Related Documents

Data Sheet i305 | d-006699-012838.pdf
available at www.duagon.com

Configure ionia™ online:



duagon AG
Riedstrasse 12
CH-8953 Dietikon
Phone +41 44 743 73 00
Fax +41 44 743 73 15

www.duagon.com

duagon®
Moving TCMS Solutions